Huawei Access Node BAS-ACC-73 Basavanagudi POP QoS Congestion Event (ALARM_4005) Resolution SOP

Node: BAS-ACC-73 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_4005

Description

This procedure addresses a warning-level QoS congestion event, specifically microbursts and interface tail drops, on the Huawei Access Node BAS-ACC-73 located at the Basavanagudi POP. The failure domain is rooted in the inherent protocol and hardware limitations of the Huawei Access Node, which may lead to QoS queue starvation under heavy traffic conditions. The Huawei Access Node utilizes a combination of scheduling algorithms and buffer management techniques to prioritize and manage incoming traffic. However, the presence of microbursts can overwhelm the queue management mechanisms, resulting in interface tail drops and subsequent QoS congestion events. This SOP outlines the steps necessary to diagnose and resolve the issue, ensuring optimal network performance and minimizing downtime.

Pre-requisite Checks

- check: Verify the Basavanagudi POP's environmental conditions, including temperature and humidity levels, to ensure they are within the specified operating range.

- check: Confirm the redundant link status for BAS-ACC-73 to ensure redundancy is functioning as expected.

- check: Verify that the Huawei Access Node BAS-ACC-73 is running the latest software version and has received all necessary firmware updates.

Potential Root Causes

Cause	Probability
Inadequate QoS configuration	High
Hardware limitations (e.g., buffer size, scheduling algorithm)	Medium
Software bugs or firmware issues	Low

Diagnostic Commands & Expected Outputs

```
Command: display qos queue, Expected Output: Queue configuration and status, Description: This command displays the QoS queue configuration, including queue sizes, scheduling algorithms, and buffer management settings. The output will help identify potential queue starvation issues.

Command: display interface brief, Expected Output: Interface statistics and status, Description: This command displays a brief
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summary of each interface, including statistics and status information. The output will help identify interface tail drops and potential congestion issues.

Command: display scheduler, Expected Output: Scheduling algorithm configuration and status, Description: This command displays the scheduling algorithm configuration, including scheduling priorities and weights. The output will help identify potential scheduling conflicts or issues.
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Resolution Steps

Step 1: Step: Verify and adjust QoS configuration, Command: qos queue-profile modify, Description: Adjust the QoS queue configuration to ensure optimal queue management and prevent queue starvation. This may involve modifying queue sizes, scheduling algorithms, or buffer management settings.

Step 2: Step: Clear interface statistics, Command: clear interface statistics, Description: Clear interface statistics to prevent false positives and ensure accurate monitoring and troubleshooting.

Step 3: Step: Verify and adjust scheduling algorithm configuration, Command: scheduler modify, Description: Verify and adjust the scheduling algorithm configuration to ensure optimal scheduling and prevent scheduling conflicts or issues.

Post-Check Verification

```
Command: display qos queue, Expected Output: Queue configuration and status, Description: Verify that the QoS queue configuration has been updated and is functioning as expected.

Command: display interface brief, Expected Output: Interface statistics and status, Description: Verify that interface tail drops have been resolved and interface statistics are within expected ranges.

Command: display scheduler, Expected Output: Scheduling algorithm configuration and status, Description: Verify that the scheduling algorithm configuration has been updated and is functioning as expected.
```

Roll-back Procedures

- Step: Restore original QoS configuration, Command: qos queue-profile restore, Description: Restore the original QoS queue configuration to prevent unintended consequences.

- Step: Restore original scheduling algorithm configuration, Command: scheduler restore, Description: Restore the original scheduling algorithm configuration to prevent unintended consequences.